

Exploring Global Developer Ecosystem: Insights from the Stack Overflow Survey

Maria Carolina
Sanabria-Salas

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OUTLINE



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EXECUTIVE SUMMARY

This project analyzes developer survey and job market data to identify current technology usage, future industry trends, compensation patterns, and developer demographics in the global software industry.



- **Market Demand:** Washington DC has the highest concentration of technology job postings.
- **High-Value Skills:** Swift, Python, and C++ are the highest-paying programming languages, with average salaries exceeding \$113,000 annually.
- **Programming Languages:** JavaScript and SQL remain the most widely used programming languages, while Go and Rust show increasing interest for future adoption.
- **Database Trends:** PostgreSQL is the leading database for both current and future use. MySQL interest declined, allowing Redis to move into the second position for future preferences.
- **Developer Demographics:** Most respondents are professionals aged 25–34 and primarily hold a Bachelor's degree.
- **Cloud Platforms:** Amazon Web Services (AWS) is the most used and desired cloud platform, followed by Microsoft Azure and Google Cloud.



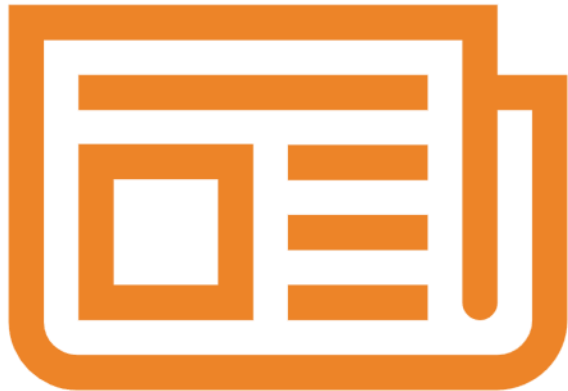
INTRODUCTION



- **Purpose:** To explore the global developer ecosystem by analyzing trends in technology usage, salaries, and demographics based on the Stack Overflow Developer Survey.
- **Target Audience:** Tech recruiters, educational institutions, developers planning their career paths, and organizations looking to align their tech stacks with market trends.
- **Value of Analysis:** Provides data-driven insights into which technologies are gaining traction and which skills offer the highest financial returns



METHODOLOGY



Data Sources

- Supplementary job postings and salary dataset (Module 1)
- Stack Overflow Developer Survey (main dataset, 18,845 rows × 114 columns)

Data Collection, Selection, & Data Wrangling

- Reduced full dataset to ~20 relevant variables related to technology usage, compensation, demographics, and job satisfaction
- Cleaned missing values and standardized formats
- Converted salary and experience fields to numeric types
- Handled multi-response fields (e.g., languages, databases) by splitting and exploding values
- Feature Engineering:
 - Created grouped variables (e.g., Employment_Group, ExperienceLevel)
 - Generated frequency tables for top technologies (Top 10 languages/databases/frameworks)
 - Applied outlier removal using IQR method for compensation analysis

Analysis & Visualization

- Python (Pandas/Seaborn/Matplotlib) was used to create bar charts, distribution plots, correlation heatmaps, and descriptive analyses.
- The IBM Cognos dashboard integrated word clouds, bubble charts, treemaps, maps, and stacked bar charts.



Results

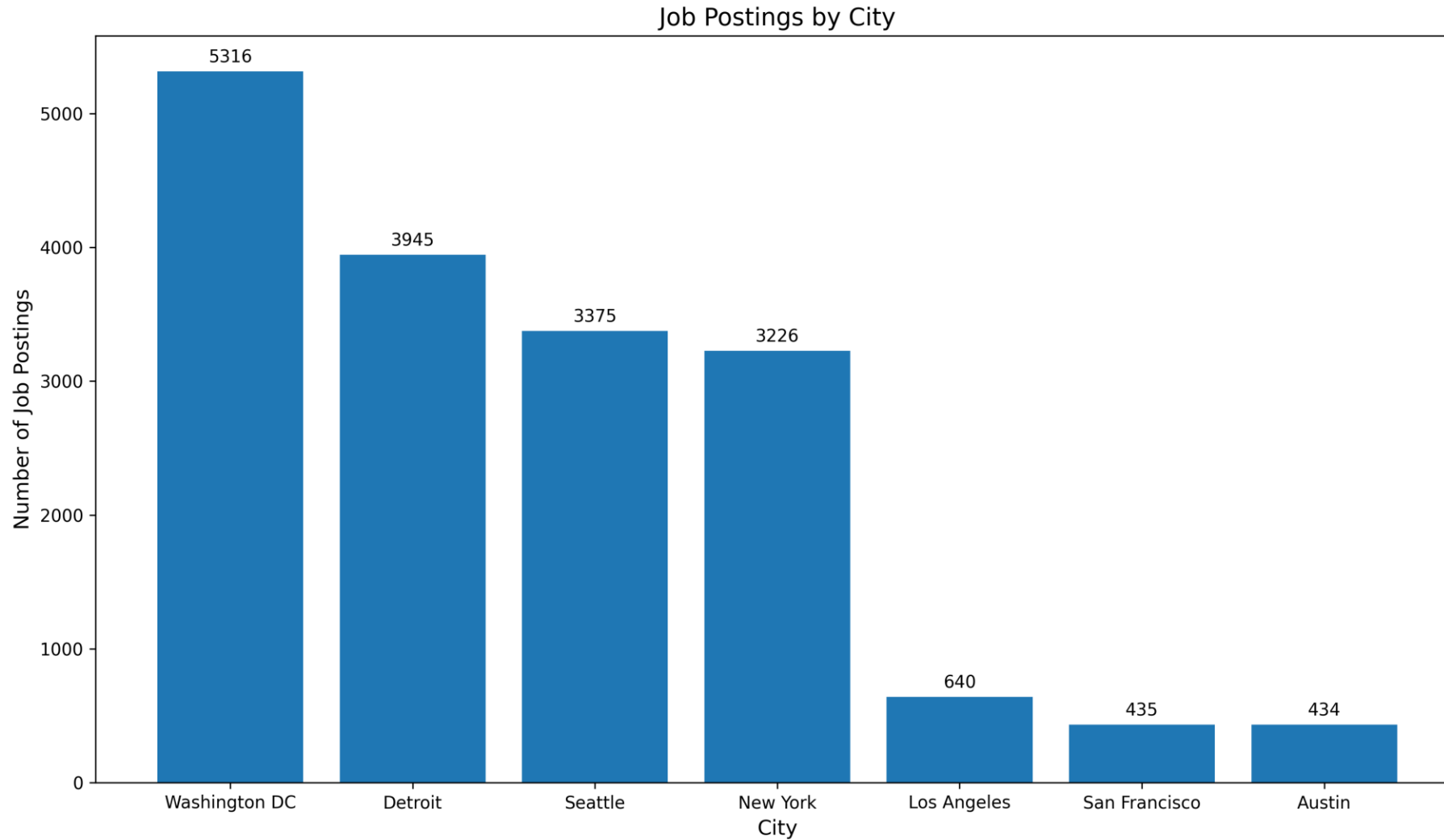
Repository:

https://github.com/carolina2401/IBM_DataAnalystCapstoneProject.git



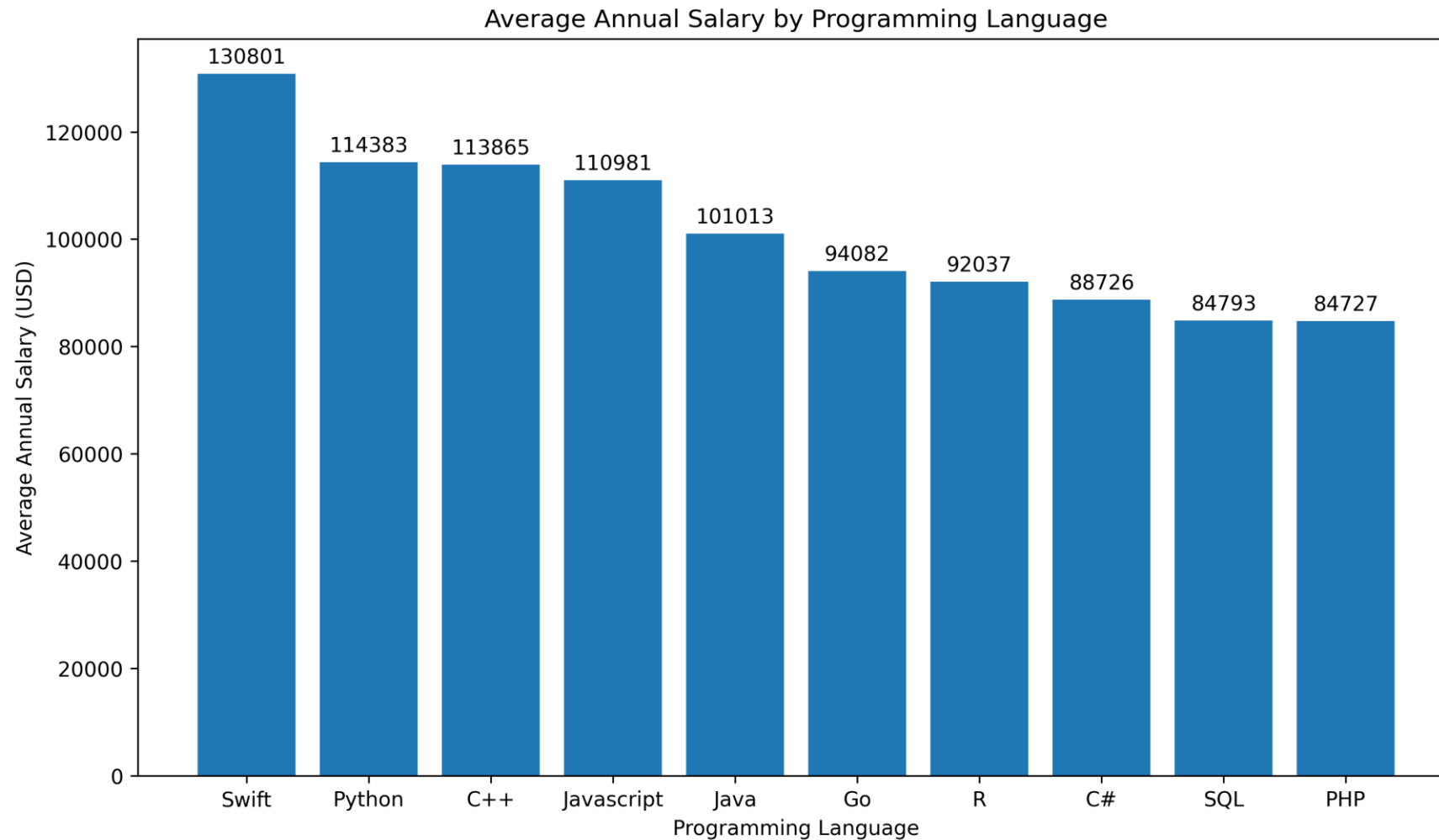
JOB POSTINGS

Washington DC leads the market with 5,316 job postings, followed by Detroit (3,945) and Seattle (3,375)



POPULAR LANGUAGES

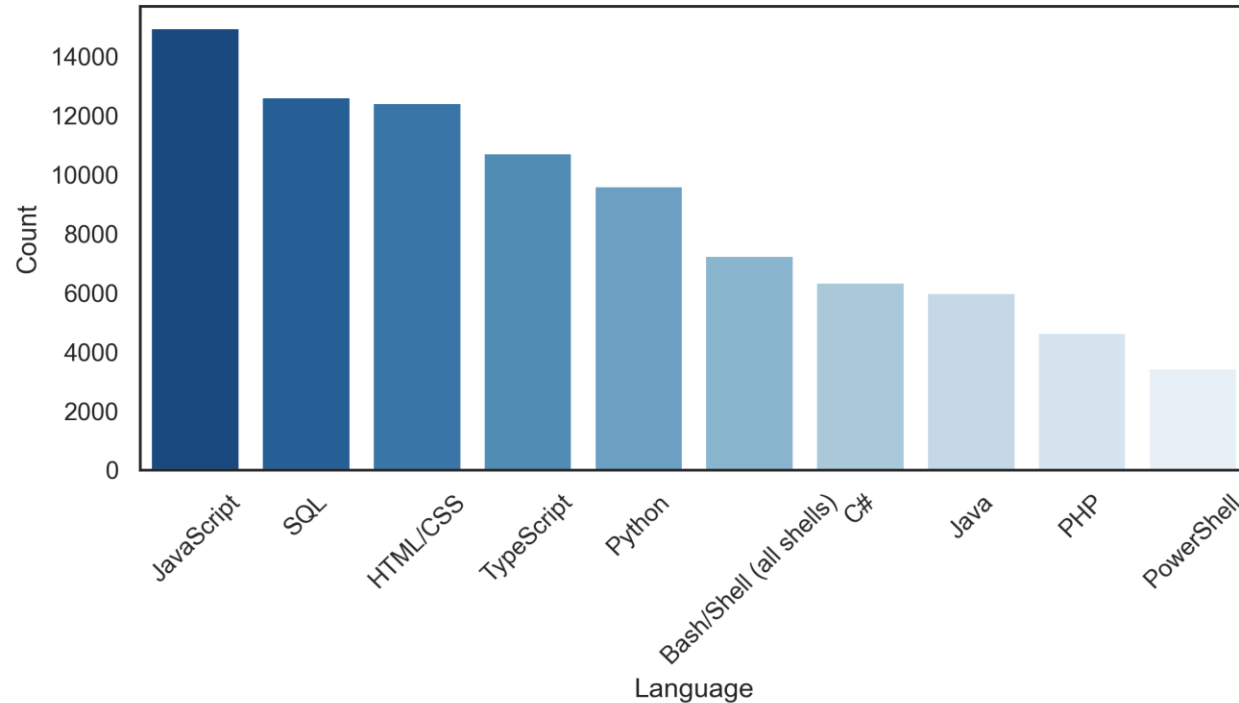
Swift, Python, and C++ are the highest-paying languages (> \$113,000)



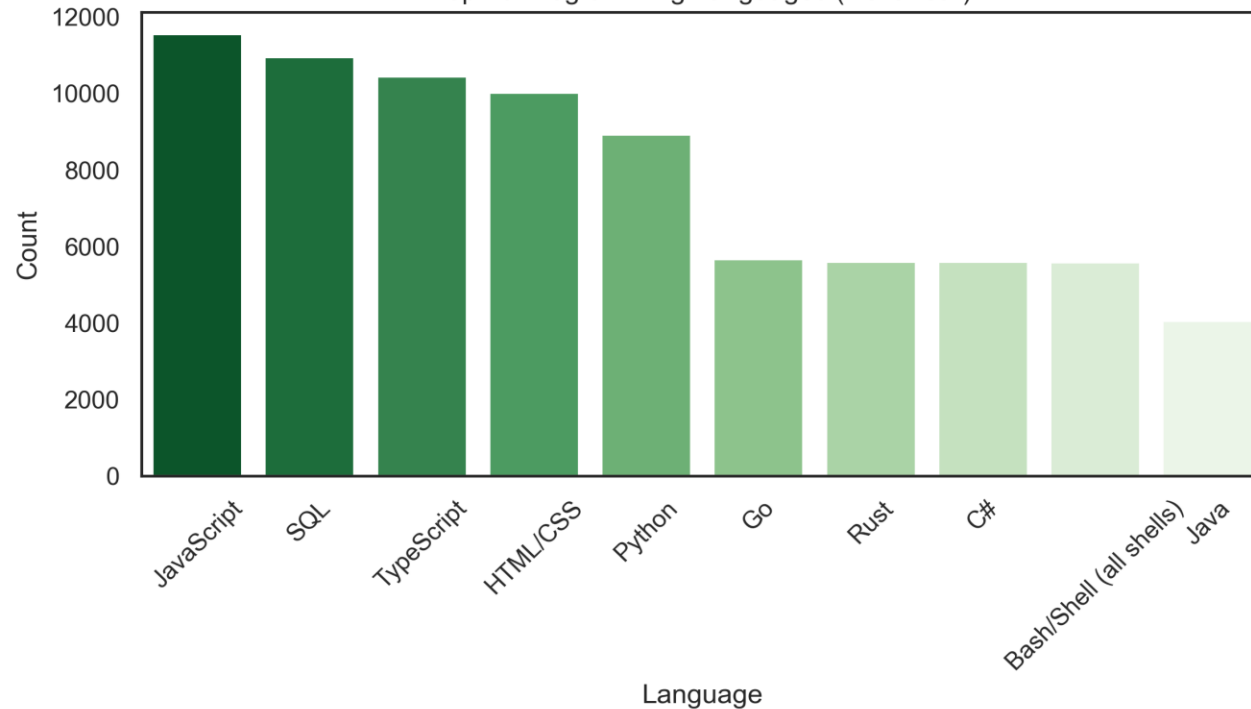
PROGRAMMING LANGUAGE TRENDS

JavaScript and **SQL** are the dominant languages for current users and are the top choice for future projects, while TypeScript, HTML/CSS, and Python continue to show robust demand

Top 10 Programming Languages (Current Year)



Top 10 Programming Languages (Next Year)



Data Module 4 - Stack Overflow Developer Survey



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript, SQL, HTML/CSS, TypeScript, and Python are leading current and future adoption
- Go and Rust show increasing interest for future adoption.

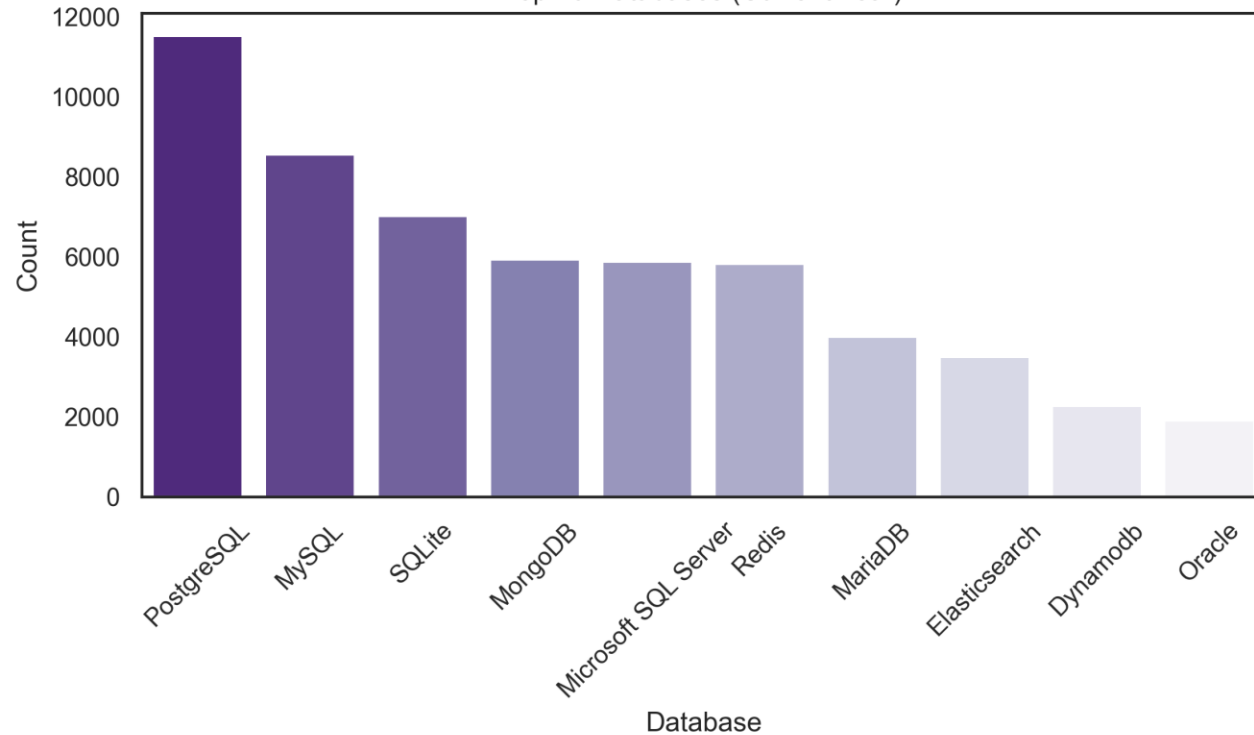
Implications

- Prioritize Python & TypeScript for future-proofing.
- Adopt Rust & Go for performance-critical and cloud-native projects.
- Phase out legacy languages like Java and PHP

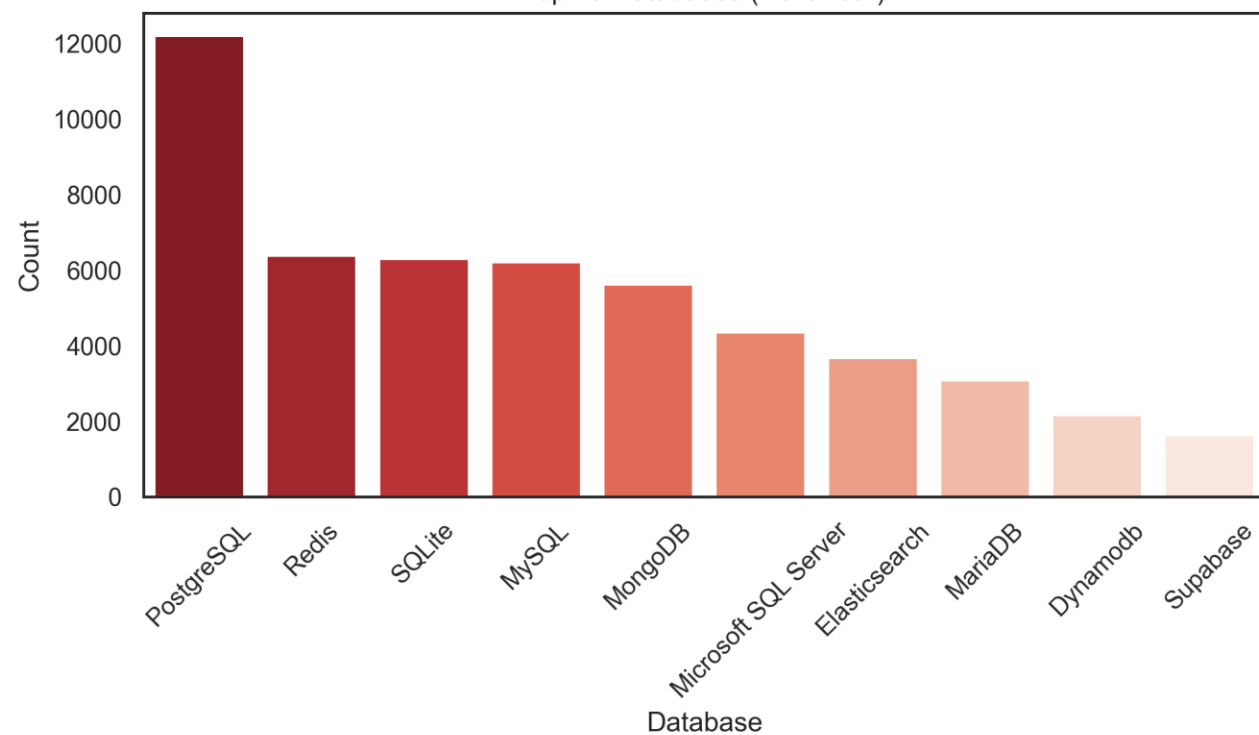
DATABASE TRENDS

PostgreSQL is the dominant database in the current year and is set to maintain its leadership as the most desired future choice, with interest growing to over 12,000 respondents

Top 10 Databases (Current Year)



Top 10 Databases (Next Year)



Data Module 4 - Stack Overflow Developer Survey



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL dominates current usage, followed by MySQL, and SQLite usage
- Redis has emerged as the second most desired database for future adoption

Implications

- Skilled up in SQL and NoSQL, as both are widely used and growing
- New projects should favor cloud-native databases and platforms
- Start managing cloud services instead of self-hosted



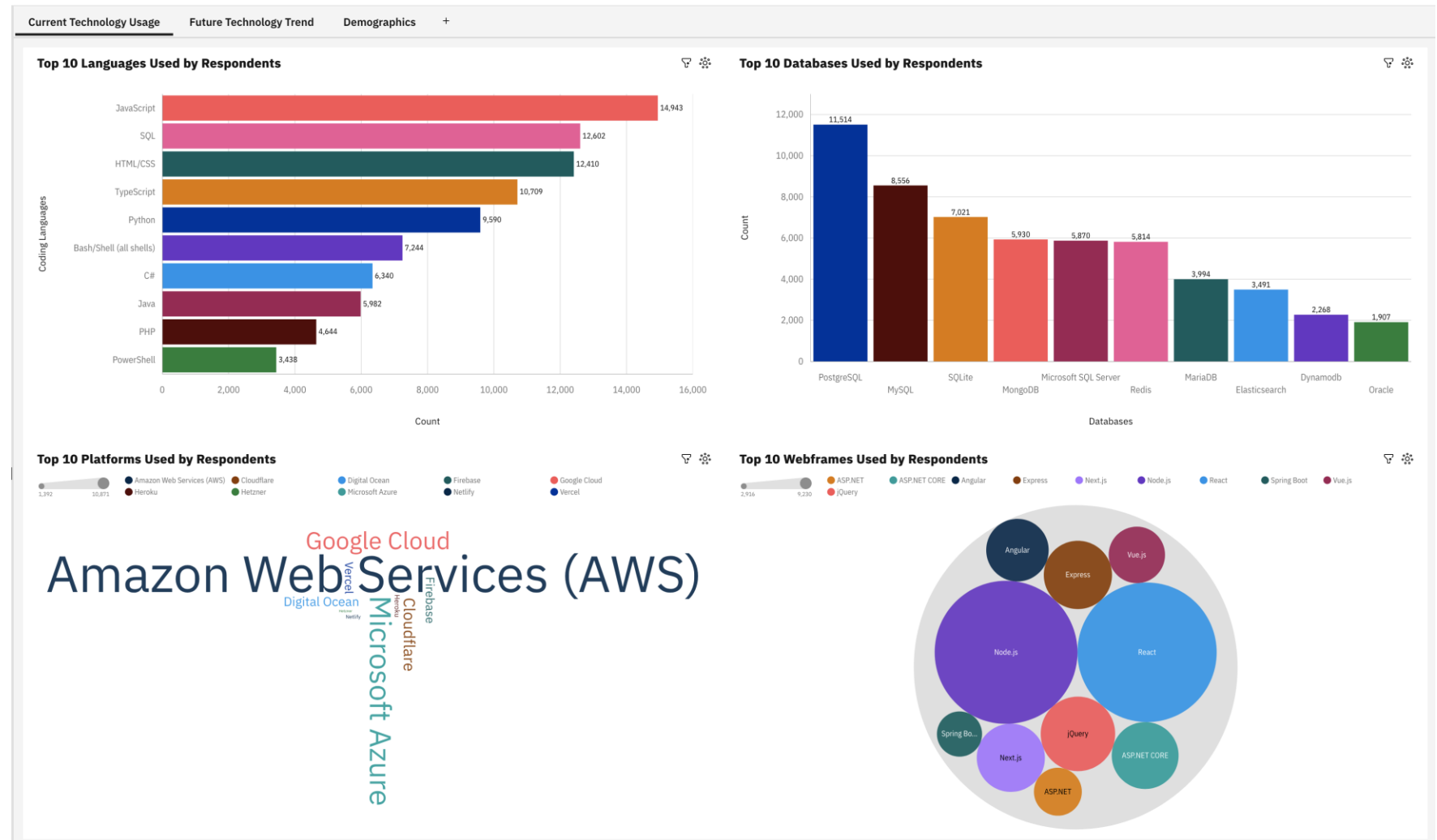
DASHBOARD



Interactive Dashboard:

[Stack-Overflow-Survey-Dashboard-IBMcognos.pdf](#)

DASHBOARD TAB 1: Current Technology Usage



Cloud Platforms:
AWS is the clear market leader, visually dominating the platform word cloud over Azure and Google Cloud

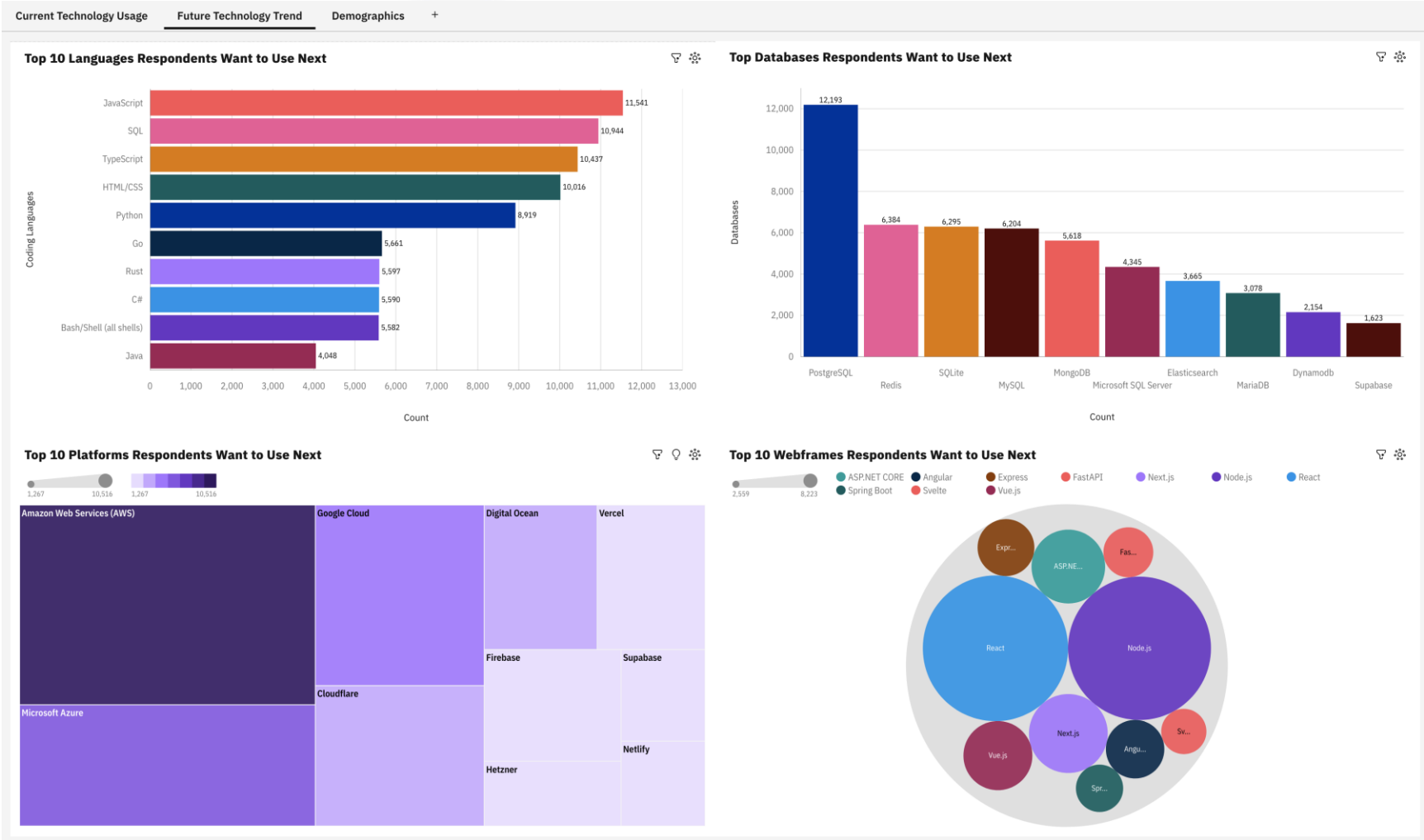
Web Frameworks:
Node.js and React are the dominant web frameworks by a significant margin



DASHBOARD TAB 2: Future Technology Trends

Platform Growth:
The treemap indicates that while **AWS** maintains its lead, there is significant future interest in Microsoft Azure and Vercel

Webframe Evolution:
React and **Node.js** remain the top desired frameworks, while Next.js and FastAPI are emerging as popular choices for future projects



DASHBOARD TAB 3: Demographics

Age Profile:

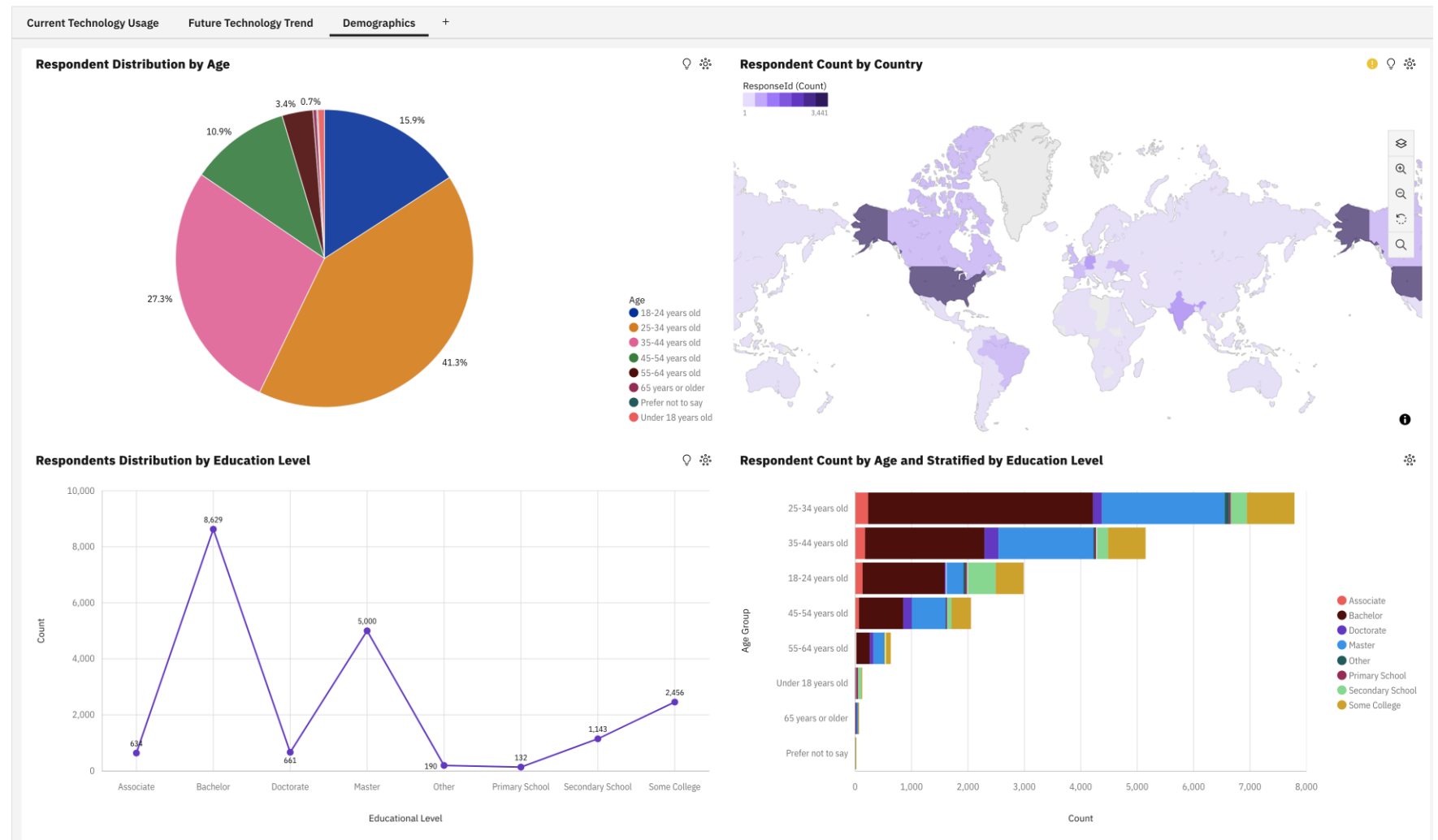
The developer population is young (41.3% of respondents are 25–34 years old)

Education:

Most respondents have a Bachelor's degree (8,629), or Master's degree (5,000)

Geographic Concentration:

The highest density of developers is found in the United States, India, and Western Europe



DISCUSSION



- A few tools dominate the entire industry — mastering them is enough to be employable.
- Developers want change but stay safe — future demand mirrors current usage with minor shifts.
- The workforce is young, educated, and geographically concentrated — talent supply is unequal but predictable.

OVERALL FINDINGS & IMPLICATIONS

Findings

- Washington DC leads developer job demand
- Swift and Python offer the highest salary potential
- JavaScript, SQL, PostgreSQL, and AWS dominate the tech ecosystem
- Developers are mainly young, university-educated professionals

Implications

- Build skills in both foundational and specialized technologies
- Shift toward cloud-native and modern typed systems
- Align hiring with young talent in major tech markets
- Standardize around widely adopted platforms like AWS and React

CONCLUSION

Market Strategy:

- Focus recruitment and training on the AWS/PostgreSQL/JavaScript ecosystem for maximum alignment.

Career Growth:

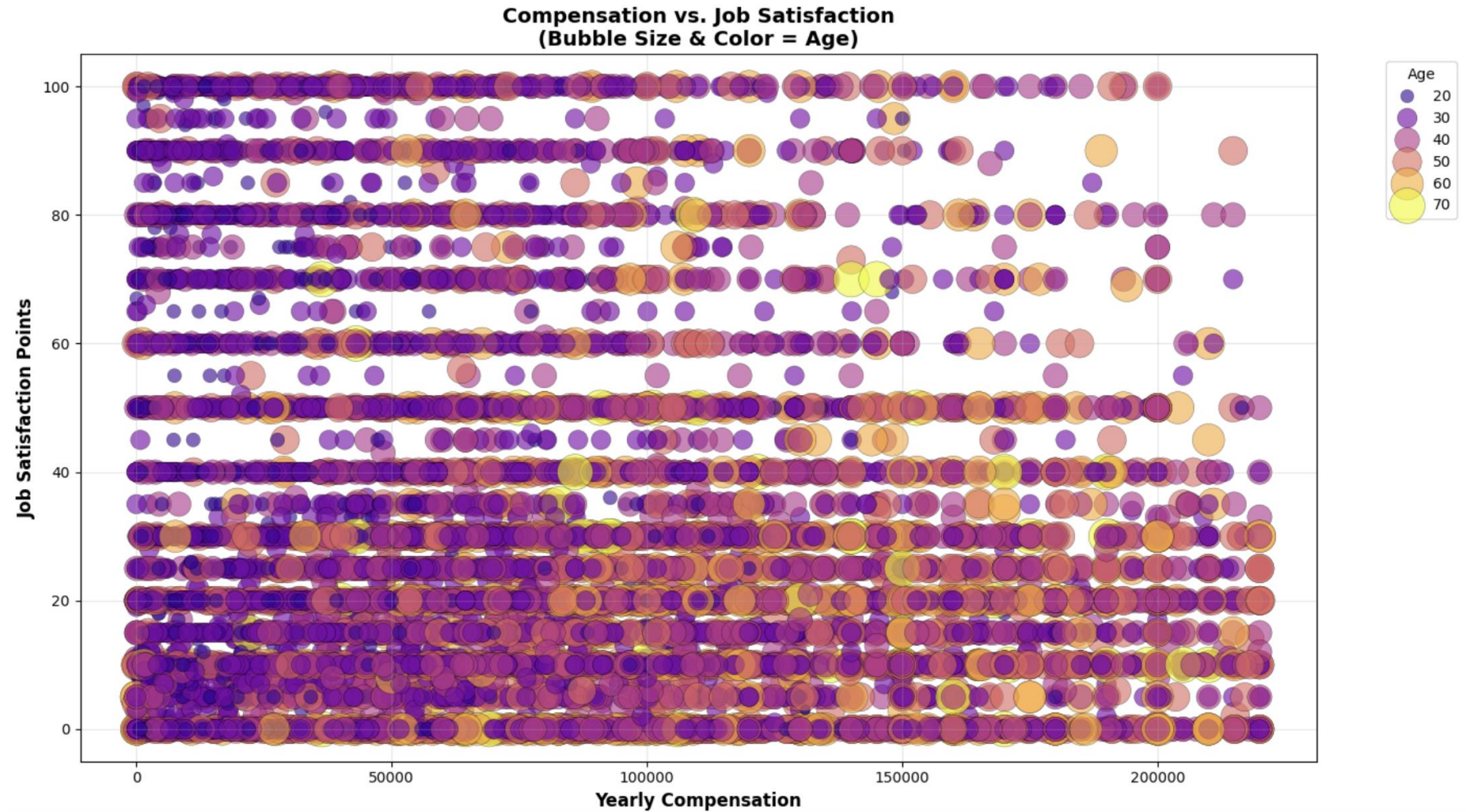
- Master high-demand core tools like Python first, then add Go or Rust to stay ahead of the next hiring wave.
- Mastering Swift or specialized cloud services offers the highest financial incentives.

Demographic Target:

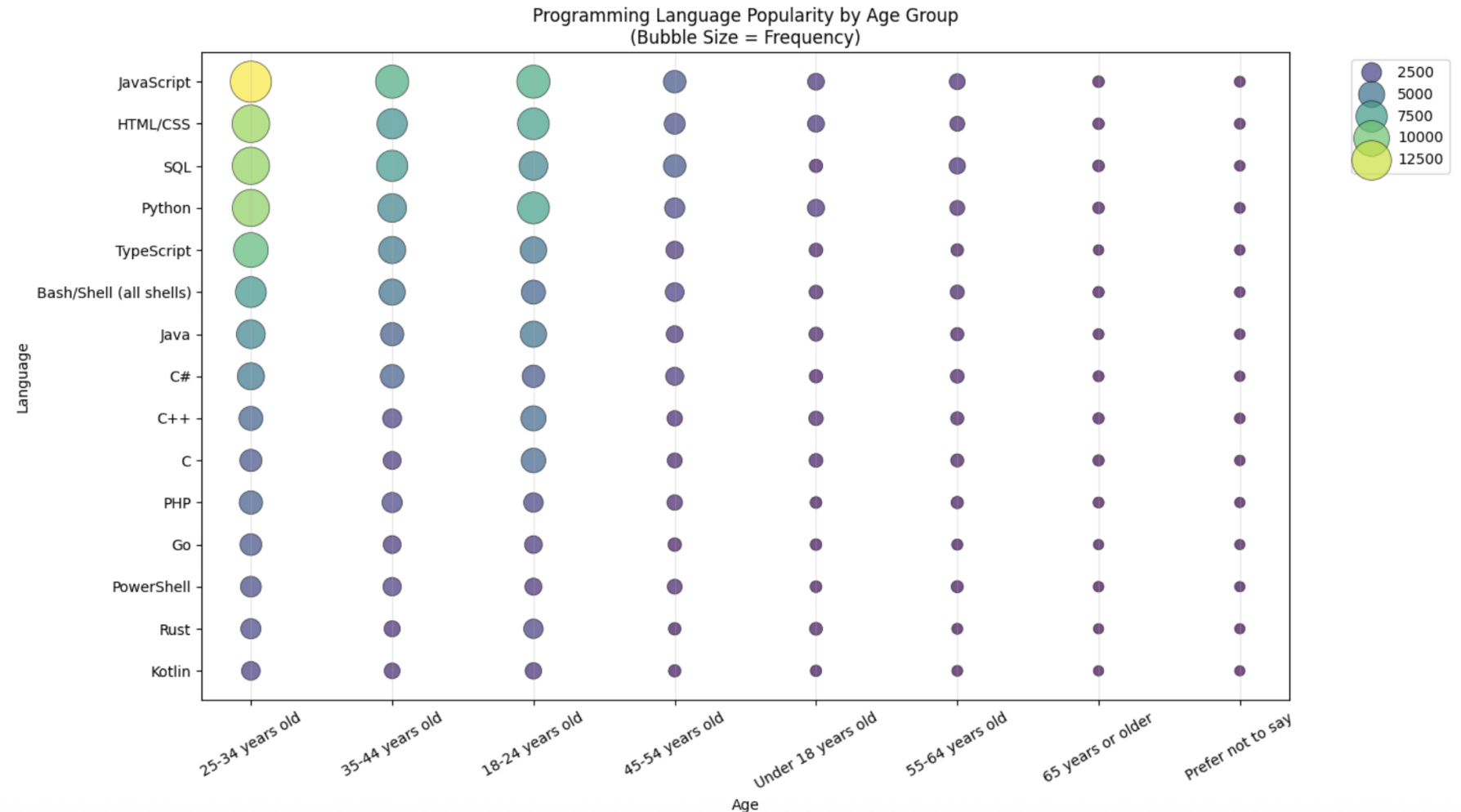
- Degree-holding professionals aged 25–34 are the largest and most active group (best audience in tech ecosystem)



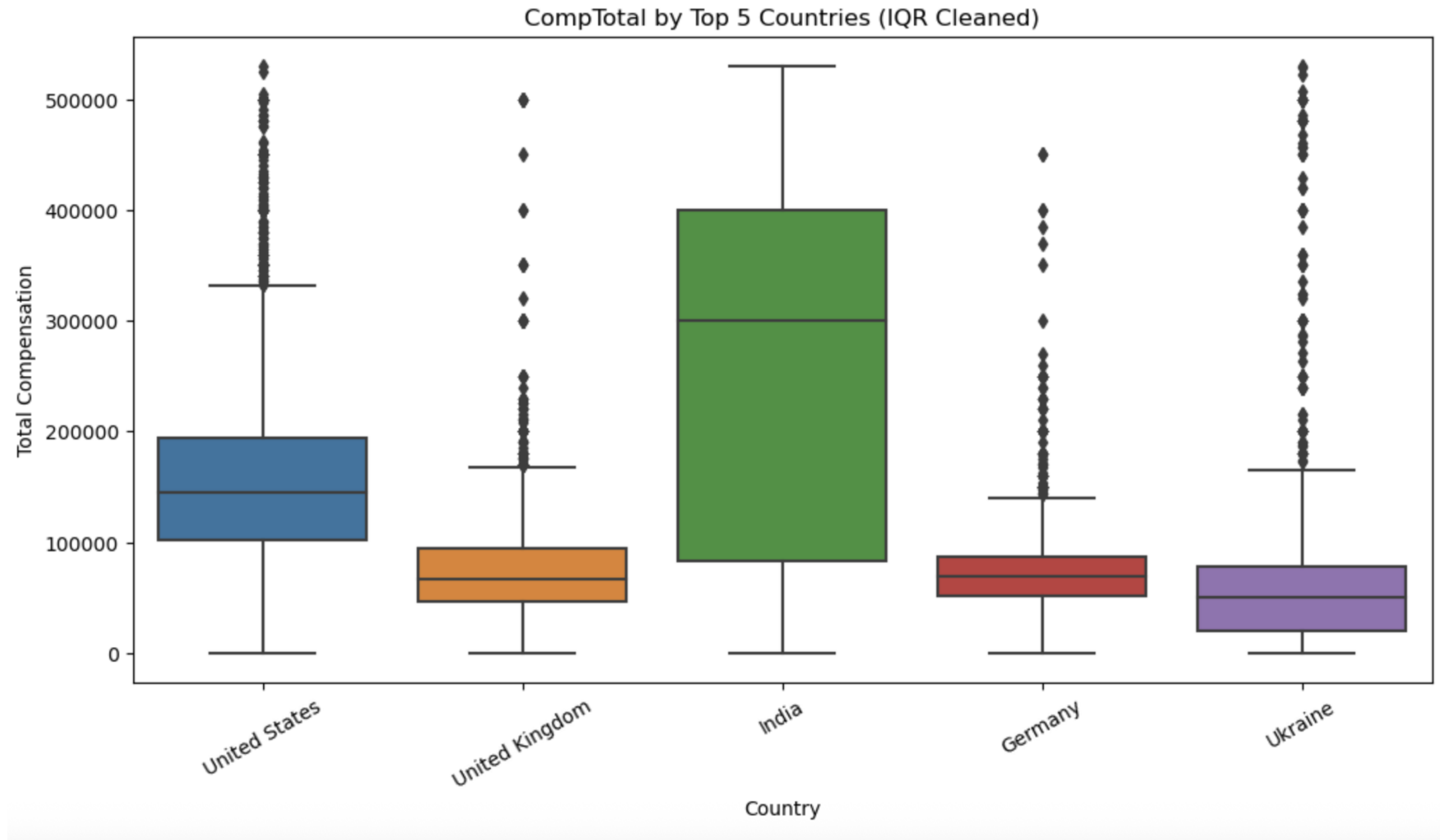
APPENDIX



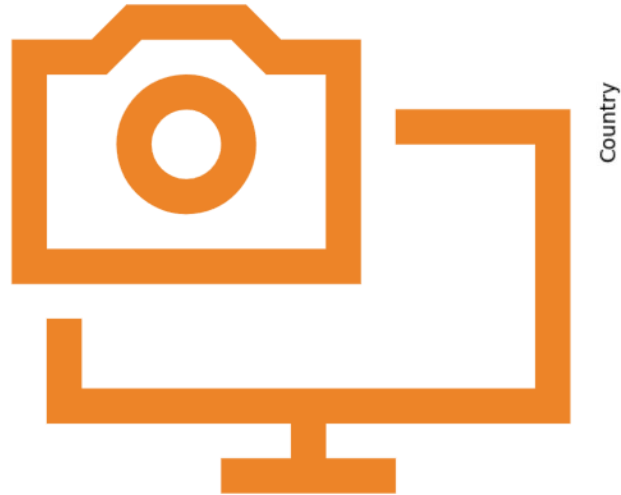
APPENDIX



APPENDIX



APPENDIX



Country

United Kingdom of Great Britain and Northern Ireland

United States of America

Programming Language Popularity by Country

